Utility

Primary Reference: Intermediate Varian

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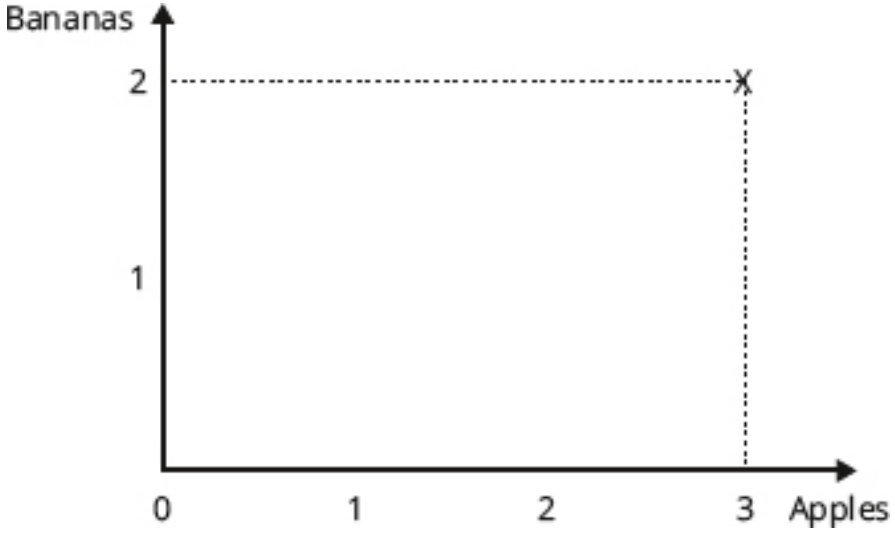
January 27th, 2026

The Baseline: Decision Making under Certainty

- We first need to understand a few things that are fundamental to economics.
- Economics, among other things, is a decision science (personally, this is my favourite part about the subject).
- We first need to understand how rationality comes in here.
- We will do so by exploring some axioms and definitions.

Preferences

- *Preference* is a relation.
- The *universe* is a set of all things that can be related.
- In economics, we are interested in preferences over consumption bundles.
- Needless to say, bundles can consist of one or multiple goods. The following bundle can be written as $\langle 3, 2 \rangle$.



Rational Preferences

- A rational preference relation R is one that is
  1. Complete.
  2. Transitive. This means that if x bears relation R to y and y bears relation R to z, x must bear relation R to z.
- Multiple relations can be complete and transitive. Consider set $\mathbb{U} = \{\text{Federer, Nadal and Djokovic}\}$.
  - Wimbledon titles: Federer (8) has more than Djokovic (7), who has more than Nadal (2). Hence, Federer has more than Nadal. This relation is complete and transitive.
  - French Open titles: Nadal (14) has more than Djokovic (3), who has more than Federer (1). Hence, Nadal has more than Federer. This relation is complete and transitive.

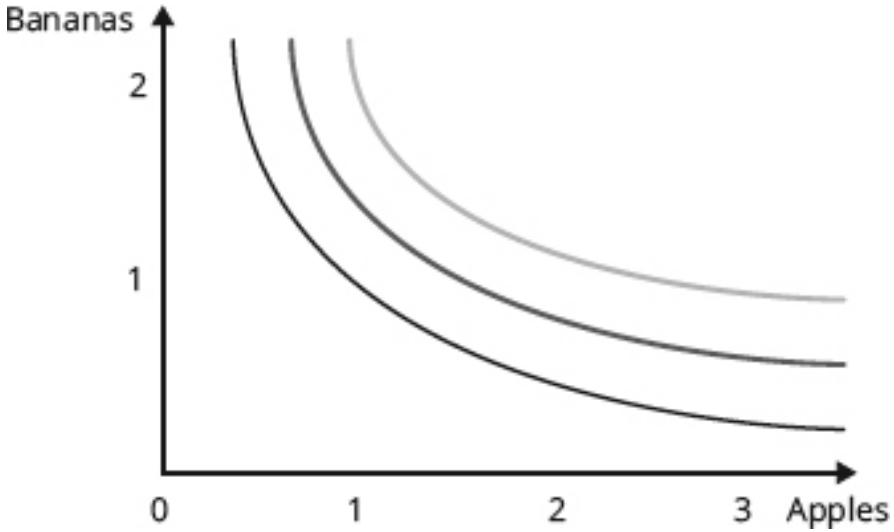
Rational Preferences

- Note the above relations are not preference relations and hence, talking about rationality wouldn't make much sense.
- However, you can imagine that different people can have different preferences over the same set of bundles.
- As long as the preferences are complete and transitive, they are rational.

Formal Definitions

- *Completeness of* $\succcurlyeq$. Either $x \succcurlyeq y$ or $y \succcurlyeq x$ (or both) (for all x,y).
- *Transitivity of* $\succcurlyeq$. If $x \succcurlyeq y$ and $y \succcurlyeq z$, then $x \succcurlyeq z$ (for all x,y,z).
- *Indifference.* $x \sim y$ if and only if $x \succcurlyeq y$ and $y \succcurlyeq x$.
- *Strict Preference.* $x \succ y$ if and only if $x \succcurlyeq y$ and it is not the case that $y \succcurlyeq x$.

The Indifference Curve



Utility

- A **utility function** associates a number with each member of the set of alternatives.
- This is where the fun starts.
- A function $u(\cdot)$ from the set of alternatives into the set of real numbers is a utility function representing the preference relation $\succcurlyeq$ just in case $x \succcurlyeq y \Leftrightarrow u(x) \geq u(y)$ (for all x and y).

How to relate IC to utility?

- Suppose we have a utility function

$$u(x_1, x_2) = x_1 x_2$$

- If we want to construct and indifference curve from this, we need a constant $u(x_1, x_2) = k$.

Indifference curves. The indifference curves $k = x_1 x_2$ for different values of k.

How to relate IC to utility?

- Remember that the utility function is not the same as the IC.
- Perfect substitutes: $ax_1 + bx_2$. Slope is $-\frac{a}{b}$.
- Perfect complements: $\min\{ax_1, bx_2\}$.
- Quasilinear preferences: $u(x_1, x_2) = v(x_1) + x_2$